SAT Math Mastery · Session 6

Percentages, Ratios and Rates

Time 3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.

Topics percent of a quantity · percent increase and decrease · successive and reverse percent changes · percent error · mixtures and weighted percents · percent chains and parameters · ratios and proportions · rates and direct variation · unit conversion, including squared and cubed units · scale and relative speed · solving with a given total

Name / Date _____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos, one expression per line; where two curves are typed, tap the grey dot at their intersection to read the answer. Where a letter such as x stands for another quantity, the step says so.

1. HARD · NO CALCULATOR NEEDED

The table below summarizes the number of students in each of four school clubs.

Club	Number of students
A	500
B	72
C	1,100
D	128
Total	1,800

The number of students in club C is $p\%$ of the number of students in club A. What is the value of p?

1. Club C is $p\%$ of club A, so $p = \dfrac{\text{club C}}{\text{club A}} \times 100$. Only these two clubs matter; ignore B, D and the total.

2. Type `1100/500*100` : it returns 220.

3. Check: type `500*220/100` : it returns 1,100, which is the club C count.

Answer: 220

2. Hard · Desmos

In a factory quality check, 40% of the items inspected in a batch are defective. Of those defective items, 45% are also mislabeled. Which of the following is closest to the percentage of items in the batch that are NOT mislabeled defective items?

(A) 55%

(B) 82% ✓

(C) 96.4%

(D) 18%

1. The share that is defective and mislabeled is 45% of 40%. Type `0.45*0.4*100`: it returns 18 percent of the batch.
2. Everything else is NOT in that group: type `100-18`: it returns 82 percent.
3. Trap check: type `100-45`: it returns 55, which is only the share of defective items that are not mislabeled; the question is about the whole batch.

Answer: (B)

3. Hard · Desmos

The population of a town increased by 10% in the first year and then decreased by 10% in the second year. If the population at the start of the first year was 5,000, what was the population at the end of the second year?

(A) 5,050

(B) 4,900

(C) 4,950 ✓

(D) 5,000

1. A 10% increase multiplies by 1.1. Type `5000*1.1`: it returns 5,500.
2. The 10% decrease is taken from the new 5,500: type `5500*0.9`: it returns 4,950 people.
3. One-line check: type `5000*1.1*0.9`: it returns 4,950, which is 50 below the start because 10% of the larger number is bigger.

Answer: (C)

4. Hard · Desmos

A renewable-energy report estimated that a new solar farm would generate 480 gigawatt-hours (GWh) of electricity in its first year. The farm actually generated 432 GWh. What is the percent error of the estimate, relative to the actual amount generated?

(A) 11.1% ✓

(B) 12.5%

(C) 9.1%

(D) 10.0%

1. Percent error divides the error by the actual amount. Type `480-432`: it returns 48 gigawatt-hours.
2. Type `48/432*100`: it returns 11.1111, which rounds to 11.1%.
3. Wrong-base check: type `48/480*100`: it returns 10, which is the tempting answer from dividing by the estimate.

Answer: (A)

5. HARD · DESMOS

The price of an item is increased by 40%, and then the new price is decreased by 25%. The final price is what percent of the original price?

(A) 115% (B) 65%

(C) 100% **(D) 105% ✓**

1. Pick a starting price of 100. A 40% increase: type `100*1.4` : it returns 140.
2. The 25% decrease is taken from the new price: type `140*0.75` : it returns 105 dollars.
3. One-line check: type `1.4*0.75*100` : it returns 105, so the final price is 105% of the original. Subtracting the percents would give 115%, a trap.

Answer: (D)

6. HARD · DESMOS

A store sells an item at 40% above its cost, then offers a 25% discount. The discounted selling price is \$210. What was the store's cost, in dollars?

1. Let x stand for the cost. Selling at 40% above cost is $1.4x$, and the 25% discount leaves 75% of that, so the discounted price is $0.75 \times 1.4x$.
2. Type `y=0.75*1.4*x` and `y=210` (zoom out). Tap the intersection: (200, 210).
3. Check: type `200*1.4*0.75` : it returns 210 dollars, the discounted price.

Answer: 200

7. HARD · DESMOS

A town's population decreased by 20% in 2023 and then increased by 15% in 2024, ending at 9,200. What was the population before the 2023 decrease?

1. Let x stand for the population before 2023. After a 20% decrease and a 15% increase it is $0.8 \times 1.15x$.
2. Type `y=0.8*1.15*x` and `y=9200` (zoom out). Tap the intersection: (10000, 9200).
3. Check: type `10000*0.8*1.15` : it returns 9,200, the ending population.

Answer: 10000

8. HARD · DESMOS

The number x is 50% greater than y, and y is 20% less than z, where $z > 0$. The number x is what percent of z?

1. Take $z = 100$. y is 20% less than z: type `100*0.8`: it returns 80.

2. x is 50% greater than y: type `80*1.5`: it returns 120.

3. One-line check: type `1.5*0.8*100`: it returns 120, so x is 120% of z. ($50 - 20 = 30$ would be wrong.)

Answer: 120

9. HARD · DESMOS

A quantity increases by 10%, then increases by 10% again, and then decreases by 20%. By what percent has the quantity decreased from its original value? (Enter as a decimal.)

1. Start with 100. First 10% increase: type `100*1.1`: it returns 110.

2. Second 10% increase, taken from 110: type `110*1.1`: it returns 121.

3. The 20% decrease: type `121*0.8`: it returns 96.8.

4. Decrease from 100: type `100-96.8`: it returns 3.2 percent.

Answer: 3.2

10. HARD · DESMOS

A rectangular banner has an area of 540 square inches. The length of the banner is 150% of its width, and the width is $c\sqrt{10}$ inches, where c is a positive constant. What is the value of c?

1. Let x stand for the width w. The length is $1.5w$, so the area is $1.5w^2 = 540$.

2. Type `y=1.5*x^2` and `y=540` (use the positive crossing). Tap the intersection: (18.9737, 540).

3. The width is $c\sqrt{10}$, so $c = \dfrac{w}{\sqrt{10}}$: type `sqrt(360)/sqrt(10)`: it returns 6.

Answer: 6

11. HARD · DESMOS

A metalworker melts together 150 g of alloy X, which is 20% nickel by mass, and m g of alloy Y, which is 50% nickel by mass. Then 40 g of pure nickel is stirred into the melt. The final alloy is 40% nickel by mass. What is the value of m?

(A) 250 (B) 460

(C) 60 ✓ (D) 100

1. Nickel in grams: alloy X gives 0.2×150, alloy Y gives $0.5m$, pure nickel gives 40. The final alloy has mass $150 + m + 40$ and is 40% nickel. Let x stand for m.
2. Type `y=0.2*150+0.5*x+40` and `y=0.4*(150+x+40)`. Tap the intersection: (60, 100).
3. Check: type `(30+0.5*60+40)/(150+60+40)`: it returns 0.4, which is 40% nickel.

Answer: (C)

12. HARD · DESMOS

The area covered by a coral colony increases by 33.1% every 3 years. If the colony grows by the same percent each year, by what percent does its area increase each year?

1. Let x stand for the yearly growth rate as a decimal. Three years of growth give $(1 + x)^3 = 1.331$.
2. Type `y=(1+x)^3` and `y=1.331`. Tap the intersection: (0.1, 1.331).
3. Convert to a percent: type `0.1*100`: it returns 10 percent per year.
4. Check: type `1.1^3`: it returns 1.331, the 33.1% growth over three years.

Answer: 10

13. HARD · DESMOS

Last year a shop sold only mugs and plates. This year the number of mugs sold rose by 30%, the number of plates sold fell by 10%, and the total number of items sold rose by 6%. Of the items sold this year, what percent were mugs, to the nearest tenth of a percent?

(A) 40.0% **(B) 49.1% ✓**

(C) 50.9% (D) 52.0%

1. Take last year's mugs as $M = 1$ and let x stand for the plates P. This year: mugs 1.3, plates $0.9x$, total $1.06(1 + x)$.
2. Type `y=1.3+0.9*x` and `y=1.06*(1+x)`. Tap the intersection: (1.5, 2.65).
3. Mugs this year are 1.3 and plates are $0.9 \times 1.5 = 1.35$. Share of mugs: type `1.3/(1.3+0.9*1.5)*100`: it returns 49.0566.
4. That rounds to 49.1%, the share of mugs this year.

Answer: (B)

14. HARD · DESMOS

In 2020, the population of city M was 20% greater than the population of city N. From 2020 to 2024, the population of city M increased by 15% and the population of city N decreased by 8%. The population of city M in 2024 was p% greater than the population of city N in 2024. What is the value of p?

1. Take city N to be 100. City M is 20% greater: type `100*1.2`: it returns 120.
2. In 2024: type `120*1.15`: it returns 138 for city M, and type `100*0.92`: it returns 92 for city N.
3. M compared with N: type `(138-92)/92*100`: it returns 50 percent greater.

Answer: 50

15. HARD · DESMOS

The positive quantities a, b, c and d satisfy the following: b is 0.6% of a, c is 5% of b, and d is 150% of c. If $b + c + d = 27$, what is the value of a?

1. Write b, c, d with a: $b = 0.006a$, $c = 0.05b$, $d = 1.5c$. Add the multipliers: type `0.006+0.05*0.006+1.5*0.05*0.006`: it returns 0.00675, so $0.00675a = 27$.
2. Type `27/0.00675`: it returns 4,000.
3. Check: type `4000*0.006`: it returns 24, which is b; type `24*0.05`: it returns 1.2, which is c; type `1.2*1.5`: it returns 1.8, which is d; type `24+1.2+1.8`: it returns 27, the required sum.

Answer: 4000

16. HARD · DESMOS

A retailer's total sales were the same in March and April. In March, online sales were p percent of the total sales and came to \$18,000. In April, online sales were $(p + 8)$ percent of the total sales and came to \$25,200. What were the retailer's total sales, in dollars, for one of these months?

1. Total sales T are equal, so the rise in online sales comes only from the 8-point rise in share. Type `25200-18000`: it returns 7,200 dollars, which is 8% of T.
2. Type `7200/0.08`: it returns 90,000 dollars.
3. Check the shares: type `18000/90000*100`: it returns 20 percent in March, type `25200/90000*100`: it returns 28 percent in April, which is 8 more.

Answer: 90000

17. HARD · DESMOS

A chemist has x liters of an acid solution that is 30% acid and mixes in 15 liters of pure acid. She then adds 25 liters of a second acid solution that is 10% acid. The final solution is 35% acid. What is the total volume, in liters, of the final solution?

(A) 125

(B) 70

(C) 85

(D) 110 ✓

1. Count the acid, with x standing for the first solution's volume: $0.3x + 15 + 0.1(25)$. The final volume is $x + 15 + 25$ and is 35% acid.
2. Type `y=0.3*x+15+0.1*25` and `y=0.35*(x+15+25)`. Tap the intersection: (70, 38.5).
3. The question asks for the total volume: type `70+15+25`: it returns 110 liters.

Answer: (D)

18. HARD · DESMOS

A library's collection consists of fiction, nonfiction and reference books. Next year the number of fiction books will increase by 20%, the number of nonfiction books will decrease by 10%, and the number of reference books will not change. After these changes, the collection will contain 1% more books than it does now, and the number of fiction books will equal the number of nonfiction books. What percent of the current collection is reference books?

(A) 65% ✓

(B) 15%

(C) 20%

(D) 35%

1. Take the current fiction books as 30. After the 20% increase: type `1.2*30`: it returns 36.
2. Nonfiction after a 10% drop equals this, so current nonfiction is type `36/0.9`: it returns 40.
3. Let x stand for the reference books. After the changes the total is $36+36+x$, and it is $1.01(30+40+x)$. Type `y=36+36+x` and `y=1.01*(30+40+x)`. Tap the intersection: (130, 202).
4. Share that is reference: type `130/(30+40+130)*100`: it returns 65 percent.

Answer: (A)

19. HARD · DESMOS

A store raises the price of a jacket by r percent. Later, it reduces the new price by $(r - 13)$ percent. After both changes, the final price is 10% higher than the original price. What is the value of r?

1. A raise of r percent multiplies the price by $1 + \frac{r}{100}$ and a cut of $(r - 13)$ percent multiplies by $1 - \frac{r - 13}{100}$. Together they give 1.1.
2. Type `y=(1+x/100)(1-(x-13)/100)` and `y=1.1` (letting x stand for r; zoom out to see both crossings).
3. Tap the intersections: $(-12, 1.1)$ and $(25, 1.1)$. A raise needs $r > 0$ (and a cut needs $r - 13 > 0$), so $r = 25$.
4. Check: type `(1+25/100)(1-(25-13)/100)`: it returns 1.1, which matches the target.

Answer: 25

20. HARD · DESMOS

Quantity Q is 60% greater than quantity P. Quantity R is 25% less than Q. Quantity R is 50% greater than quantity S. P is what percent greater than S?

1. Start from $S = 100$. R is 50% greater than S: type `100*1.5` : it returns 150.
2. R is 25% less than Q, so $R = 0.75Q$: type `150/0.75` : it returns 200, which is Q.
3. Q is 60% greater than P: type `200/1.6` : it returns 125, which is P.
4. P compared with S: type `(125-100)/100*100` : it returns 25 percent greater.

Answer: 25

21. HARD · DESMOS

An 84-gram sample of an alloy contains only copper, zinc and tin. The mass of the copper is 5 grams more than twice the mass of the zinc. The mass of the tin is half the combined mass of the copper and the zinc. What is the mass, in grams, of the tin in the sample?

1. Let x stand for the copper-plus-zinc mass s. Tin is half of it, so the whole sample is $x + \frac{x}{2} = 84$.
2. Type `y=1.5*x` and `y=84` . Tap the intersection: (56, 84).
3. Tin is half of that: type `56/2` : it returns 28 grams.
4. Check with the other clue: type `(56-5)/3` : it returns 17, which is the zinc; type `2*17+5` : it returns 39, which is the copper; type `39+17+28` : it returns 84, which is the total.

Answer: 28

22. HARD · DESMOS

For a construction project, a concrete mixture is made from cement, sand, and gravel in the ratio 1 : 2 : 3 by weight. A batching plant produces gravel at a constant rate of 6 metric tons per hour. At this rate, how many hours will it take the plant to produce 90 metric tons of the mixture?

1. The ratio 1 : 2 : 3 has total parts: type `1+2+3` : it returns 6.
2. Gravel is 3 of the 6 parts and is made at 6 tons per hour, so the whole mixture is made at type `6/(3/6)` : it returns 12 tons per hour.
3. Time for 90 tons: type `90/12` : it returns 7.5 hours.

Answer: 7.5

23. HARD · DESMOS

Car A travels 240 kilometers in 3 hours at a constant speed, and Car B travels 350 kilometers in 4.5 hours at a constant speed. Which of the following is closest to how much faster, in kilometers per hour, Car A's speed is compared with Car B's speed?

(A) 0.9

(B) 2.2 ✓

(C) 4.4

(D) 10.0

1. Speed is distance divided by time. Car A: type `240/3`: it returns 80 kilometers per hour.
2. Car B: type `350/4.5`: it returns 77.7778 kilometers per hour.
3. How much faster A is: type `240/3-350/4.5`: it returns 2.2222, which is closest to 2.2.

Answer: (B)

24. HARD · NO CALCULATOR NEEDED

A metal block has a mass of 540 grams and a volume of 60 cubic centimeters. What is the density of the block, in kilograms per cubic meter? (1 gram per cubic centimeter = 1,000 kilograms per cubic meter.)

(A) 90

(B) 900

(C) 9,000 ✓

(D) 9

1. Density is mass divided by volume. Type `540/60`: it returns 9 grams per cubic centimeter.
2. Use the given conversion, 1,000 kilograms per cubic meter for each gram per cubic centimeter: type `9*1000`: it returns 9,000 kilograms per cubic meter.
3. One-line check: type `540/60*1000`: it returns 9,000.

Answer: (C)

25. HARD · DESMOS

A water treatment plant produces fresh water at a constant rate of 1,250,000 liters per hour. At this rate, how many cubic meters of water does the plant produce in one day? (1 cubic meter = 1,000 liters.)

1. One day has 24 hours, so multiply the hourly rate by 24. Type `1250000*24`: it returns 30,000,000 liters.
2. There are 1,000 liters in a cubic meter, so divide by 1,000. Type `30000000/1000`: it returns 30,000 cubic meters.
3. Check: 1,250,000 liters per hour is 1,250 cubic meters per hour, and $1{,}250 \times 24 = 30{,}000$.

Answer: 30000

26. HARD · NO CALCULATOR NEEDED

The value of y varies directly with x. When $x = 9$, $y = 45$. What is the value of y when $x = 14$?

(A) 70 ✓ (B) 77

(C) 56 (D) 63

1. Direct variation: $y = kx$. Find k: type `45/9`: it returns 5.

2. Now y when $x = 14$: type `5*14`: it returns 70.

3. Graph check: type `y=5x` and tap $x = 14$: the point is (14, 70).

Answer: (A)

27. HARD · NO CALCULATOR NEEDED

A bus rapid transit line carries passengers at a constant average rate. The line carries 15,000 passengers over a 6-hour morning service window. At this same constant rate, which of the following is closest to the number of passengers carried during a 2.5-hour afternoon service window?

(A) 7,500 (B) 3,750

(C) 5,000 **(D) 6,250 ✓**

1. Rate: type `15000/6`: it returns 2,500 passengers per hour.

2. For 2.5 hours: type `2500*2.5`: it returns 6,250 passengers.

3. One-line check: type `15000*2.5/6`: it returns 6,250.

Answer: (D)

28. HARD · DESMOS

Two cars are racing at constant speeds around a circular track. Car A requires 12 seconds to travel once around the track, and Car B requires 20 seconds to travel once around the track. If Car A passes Car B at a certain moment, how many seconds will elapse before Car A next passes Car B?

1. Count laps. Car A gains on Car B at $\frac{1}{12} - \frac{1}{20}$ lap per second: type `1/12-1/20`: it returns 0.033333. (That is $\frac{1}{30}$.)

2. One lap of gain takes type `1/(1/12-1/20)`: it returns 30 seconds.

3. Check: type `30/12`: it returns 2.5 laps for A and type `30/20`: it returns 1.5 laps for B: A is exactly one lap ahead.

Answer: 30

29. HARD · NO CALCULATOR NEEDED

A pump fills a rectangular tank at a rate of 0.8 cubic meter per minute. The tank is 12 meters long, 5 meters wide and 2 meters deep. How many hours does it take to fill the tank completely? (Enter as a decimal.)

1. Volume of the tank: type `12*5*2`: it returns 120 cubic meters.
2. Time at 0.8 cubic meter per minute: type `120/0.8`: it returns 150 minutes.
3. In hours: type `150/60`: it returns 2.5.

Answer: 2.5

30. HARD · DESMOS

A test vehicle accelerates at a constant 22 feet per second squared. Which of the following is equal to this acceleration in miles per minute squared? (1 mile = 5280 feet)

(A) $\frac{1}{4}$ (B) $\frac{1}{240}$

(C) 15 ✓ (D) 79,200

1. Feet to miles means dividing by 5,280: type `22/5280`: it returns 0.004167 miles per second squared.
2. Seconds squared to minutes squared means multiplying by 60^2: type `22/5280*60^2`: it returns 15 miles per minute squared.
3. Check the slips: type `22*3600`: it returns 79,200 (forgot to divide by 5,280) and type `22/5280`: it returns 0.004167. Neither is the answer.

Answer: (C)

31. HARD · DESMOS

A car's fuel economy is 40 miles per gallon. Use 1 mile = 1.6 kilometers and 1 gallon = 3.8 liters. How many liters of fuel does the car need to drive 304 kilometers?

1. Work in kilometers: type `40*1.6`: it returns 64 kilometers per gallon.
2. Gallons for 304 kilometers: type `304/64`: it returns 4.75 gallons.
3. Liters: type `4.75*3.8`: it returns 18.05 liters.

Answer: 18.05

32. HARD · DESMOS

A region has a population density of 1,250 people per square kilometer. Using 1 mile = 1.6 kilometers, how many people are there per square mile?

1. A square mile is a square that is 1 mile on each side, so it measures 1.6 km by 1.6 km. Type `1.6^2`: it returns 2.56 square kilometers.
2. Each square kilometer holds 1,250 people, so one square mile holds type `1250*2.56`: it returns 3,200 people.
3. Squared unit, so use 1.6^2, not 1.6. Multiplying by only 1.6 would give 2,000.

Answer: 3200

33. HARD · DESMOS

A table has 4 legs, each with the same ground-contact area of A square centimeters. With a box on it, the table presses on the floor with a pressure of 12 newtons per square centimeter on each leg. A 120-newton weight is then added to the box, and the table is tilted so that only 3 legs touch the floor. The pressure on each of those 3 legs is now 24 newtons per square centimeter. (Pressure is force divided by contact area, and the load is shared equally by the legs touching the floor.) What is the value of A?

1. Pressure × area is force. Let x stand for A. At first the weight is $4 \times 12x = 48x$. After adding 120 it is $48x + 120$, shared by 3 legs that each carry $24x$, so $\dfrac{48x + 120}{3} = 24x$.
2. Type `y=(48*x+120)/3` and `y=24*x` (zoom out). Tap the intersection: (5, 120).
3. Check the pressure: type `(48*5+120)/3/5`: it returns 24 newtons per square centimeter on each of 3 legs.

Answer: 5

34. HARD · DESMOS

A box contains only red, green and blue flash drives, 140 in all. The ratio of red drives to green drives is 7 to 4. The ratio of blue drives to drives that are not blue is 3 to 11. How many more red drives than blue drives are in the box?

1. Blue to not-blue is 3 : 11, so blue is $\dfrac{3}{14}$ of the drives: type `140*3/14`: it returns 30, the blue drives.
2. The rest: type `140-30`: it returns 110 not blue, split 7 : 4 into red and green. type `110*7/11`: it returns 70 red.
3. Red minus blue: type `70-30`: it returns 40.

Answer: 40

35. HARD · DESMOS

A site plan was drawn so that 1 centimeter on the plan represents 40 meters of ground. A copier then enlarged the plan to 125% of its size. On the enlarged copy, the parcel appears as a rectangle measuring 7.5 centimeters by 5 centimeters. What is the actual area of the parcel, in hectares? (1 hectare = 10,000 square meters.)

(A) 384

(B) 3.84 ✓

(C) 6

(D) 38.4

1. The copy is 125% of the plan, so every length on the copy is 1.25 times its length on the plan. Thus 1.25 cm on the copy is 40 m of ground, and 1 cm on the copy is: type `40/1.25`: it returns 32 meters.
2. The parcel measures 7.5 cm by 5 cm on the copy. Its length: type `7.5*32`: it returns 240 meters. Its width: type `5*32`: it returns 160 meters.
3. Area: type `240*160`: it returns 38,400 square meters. Hectares: type `38400/10000`: it returns 3.84 hectares.
4. Slip to avoid: using 1 cm = 40 m without allowing for the enlargement gives 6 hectares.

Answer: (B)

PART B · SELF-ASSESSMENT — 22 PROBLEMS

36. MEDIUM · NO CALCULATOR NEEDED

The number m is 24% greater than 75. If m is the product of 75 and r, what is the value of r?

1. "24% greater than 75" means $m = 75 + 0.24(75)$: type `75+0.24*75` : it returns 93.

2. $m = 75r$, so $r = \dfrac{m}{75}$: type `93/75` : it returns 1.24.

3. Check: type `75*1.24` : it returns 93, which is m. (0.24 is only the increase.)

Answer: 1.24

37. MEDIUM · DESMOS

A household's monthly electricity bill was \$340.00. After a rate adjustment, the bill is now \$378.20. To the nearest tenth of a percent, by what percent did the amount of the bill increase?

(A) 11.9%

(B) 12.4%

(C) 10.1%

(D) 11.2% ✓

1. Increase in the bill: type `378.2-340` : it returns 38.2 dollars.

2. Percent increase over the old bill: type `38.2/340*100` : it returns 11.2353, which is 11.2% to the nearest tenth.

3. Wrong-base check: type `38.2/378.2*100` : it returns 10.1005, which comes from dividing by the new bill.

Answer: (D)

38. MEDIUM · DESMOS

Marcus purchased a car for \$90,000. He estimates that each year the value of the car will depreciate by 15% of the previous year's estimated value. What is the estimated value of the car exactly 2 years after Marcus purchased it?

(A) \$65,025 ✓

(B) \$76,500

(C) \$61,200

(D) \$63,750

1. Each year the value is multiplied by 0.85. After one year: type `90000*0.85` : it returns 76,500 dollars.

2. After the second year: type `76500*0.85` : it returns 65,025 dollars.

3. One-line check: type `90000*0.85^2` : it returns 65,025.

Answer: (A)

39. Medium · Desmos

A school's enrollment decreased from 720 students to 640 students over one year. What is the percent decrease in enrollment?

(A) 10%

(B) 11.1% ✓

(C) 12.5%

(D) 14.3%

1. Decrease in enrollment: type `720-640`: it returns 80 students.

2. Percent decrease uses the starting value: type `80/720*100`: it returns 11.1111, about 11.1%.

3. Wrong-base check: type `80/640*100`: it returns 12.5, which comes from dividing by the new value.

Answer: (B)

40. Medium · Desmos

A laptop has a listed price of \$3,200. A store offers a 15% discount, and then a 15% sales tax is added to the discounted price. What is the final price of the laptop, in dollars?

(A) 2,976

(B) 3,008

(C) 3,128 ✓

(D) 3,200

1. Pay 85% after the discount: type `3200*0.85`: it returns 2,720 dollars.

2. Add 15% tax to that price: type `2720*1.15`: it returns 3,128 dollars.

3. One-line check: type `0.85*1.15`: it returns 0.9775, which is the overall multiplier, so the price is below 3,200.

Answer: (C)

41. Medium · No calculator needed

After a 20% discount, the price of a jacket is \$256. What was the original price, in dollars, of the jacket before the discount?

1. After a 20% discount you pay 80% of the original, so $0.8p = 256$: type `256/0.8`: it returns 320 dollars.

2. Check: type `320*0.8`: it returns 256, the sale price.

3. Trap: adding 20% to 256 gives type `256*1.2`: it returns 307.2, which is wrong.

Answer: 320

42. Medium · Desmos

A survey of 850 eleventh-grade students found that 34% plan to take the SAT this year. Which of the following is closest to the number of students who plan to take the SAT?

(A) 289 ✓ (B) 306

(C) 255 (D) 272

1. Find 34% of 850. type `0.34*850`: it returns 289 students.
2. Check by pieces: type `850*0.1*3`: it returns 255, which is 30%, type `850*0.04`: it returns 34, which is 4%.
3. Sanity check: type `850/3`: it returns 283.3333, one third of 850, is a little less than the answer, as expected for 34%.

Answer: (A)

43. Medium · Desmos

If 30 percent of 40 percent of x is equal to 50 percent of y minus 20 percent of y, what is the value of x in terms of y?

(A) $x = 3y$ (B) $x = 5y$

(C) $x = 2y$ **(D) $x = 2.5y$ ✓**

1. Let $y = 1$ and x stand for the unknown coefficient. The left side is $0.3 \times 0.4x$ and the right side is $0.5 - 0.2$.
2. Type `y=0.3*0.4*x` and `y=0.5-0.2`. Tap the intersection: (2.5, 0.3).
3. So $x = 2.5y$. Check with $y = 2$, $x = 5$: type `0.3*0.4*5`: it returns 0.6 on the left and type `0.5*2-0.2*2`: it returns 0.6 on the right.

Answer: (D)

44. Medium · Desmos

The mass of object A is k grams. The mass of object B is 130% of the mass of object A. If the mass of object B is 66 grams greater than the mass of object A, what is the value of k?

1. Let x stand for k. Object B is $1.3x$ and is 66 more than A, so $1.3x = x + 66$.
2. Type `y=1.3*x` and `y=x+66` (zoom out). Tap the intersection: (220, 286).
3. Check: type `1.3*220-220`: it returns 66 grams more, as required.

Answer: 220

45. Medium · Desmos

A lab combines 40 grams of a solution that is 0.3% sodium chloride by mass with 60 grams of a solution that is 0.05% sodium chloride by mass. What is the percent of sodium chloride by mass in the combined 100 grams of mixture?

1. Find the grams of sodium chloride in each solution. First: type `0.003*40`: it returns 0.12 grams (that is 0.3% of 40).
2. Second: type `0.0005*60`: it returns 0.03 grams (that is 0.05% of 60).
3. Together: $0.12 + 0.03 = 0.15$ grams of salt in 100 grams of mixture, so the mixture is 0.15% sodium chloride. (Averaging 0.3 and 0.05 to get 0.175 ignores that the amounts differ.)

Answer: 0.15

46. Medium · Desmos

A car consumes fuel at a rate of 6.5 liters per 100 kilometers. At this rate, how many liters of fuel are needed to travel 340 kilometers?

(A) 20.4 (B) 21.0

(C) 22.1 ✓ (D) 23.9

1. Rate per kilometer: type `6.5/100`: it returns 0.065 liter per kilometer.
2. For 340 kilometers: type `6.5/100*340`: it returns 22.1 liters.
3. Check by scaling: type `6.5*3.4`: it returns 22.1, the same amount.

Answer: (C)

47. Medium · No calculator needed

A recipe for granola bars requires oats, honey, and raisins in the ratio 5 : 2 : 3 by weight. If a bakery uses 18 kilograms of raisins to prepare a large batch, how many kilograms of oats are used?

(A) 24 **(B) 30 ✓**

(C) 36 (D) 12

1. Raisins are 3 parts and weigh 18 kilograms: type `18/3`: it returns 6 kilograms per part.
2. Oats are 5 parts: type `5*6`: it returns 30 kilograms.
3. Check the ratio oats to raisins: type `30/18`: it returns 1.666667, which equals type `5/3`: it returns 1.666667.

Answer: (B)

48. Medium · No calculator needed

A car travels at a constant speed of 90 kilometers per hour. What is this speed in meters per second?

(A) 32.4 (B) 324

(C) 15 **(D) 25 ✓**

1. Meters per hour is 90 × 1,000, and there are 3,600 seconds in an hour: type `90*1000/3600`: it returns 25 meters per second.
2. Shortcut check: type `90/3.6`: it returns 25.
3. Reverse check: type `25*3.6`: it returns 90 kilometers per hour.

Answer: (D)

49. Medium · No calculator needed

A recipe that serves 8 people uses 3 cups of orange juice and 5 cups of water. If the same ratio of ingredients is used to make a batch that serves 24 people, how many cups of orange juice are needed?

(A) 9 ✓ (B) 15

(C) 6 (D) 7.5

1. The new batch is type `24/8`: it returns 3 times as large.
2. Orange juice: type `3*3`: it returns 9 cups.
3. Check that the ratio is kept: type `9/(9+15)`: it returns 0.375, which equals type `3/8`: it returns 0.375. (The 15 is the water.)

Answer: (A)

50. Medium · No calculator needed

A factory produces cardboard boxes at a constant rate of 480 boxes every 8 hours. At this rate, how many boxes does the factory produce in 5.5 hours?

1. Unit rate: type `480/8`: it returns 60 boxes per hour.
2. For 5.5 hours: type `60*5.5`: it returns 330 boxes.
3. One-line check: type `480*5.5/8`: it returns 330.

Answer: 330

51. Medium · No calculator needed

An architect built a scale model of a planned tower, where every 1 centimeter on the model represents 2.5 meters of the actual tower. If the model is 18 centimeters tall, what is the height, in meters, of the actual tower?

(A) 36 **(B) 45 ✓**

(C) 50 (D) 20

1. Each centimeter on the model is 2.5 meters: type `18*2.5`: it returns 45 meters.
2. Check backward: type `45/2.5`: it returns 18 centimeters, the model height.
3. Pieces check: type `10*2.5+8*2.5`: it returns 45.

Answer: (B)

52. Medium · No calculator needed

A saline solution is mixed so that the ratio of salt to water is 2 to 15. If a container holds 51 liters of the solution in total, how many liters of salt are in the container?

(A) 12 (B) 3

(C) 6 ✓ (D) 9

1. The ratio 2 : 15 has type `2+15` : it returns 17 parts in all.
2. One part: type `51/17` : it returns 3 liters.
3. Salt is 2 parts: type `2*3` : it returns 6 liters.
4. Check in one line: type `51*2/17` : it returns 6.

Answer: (C)

53. Medium · Desmos

A gift box contains almonds and cashews, in a ratio of 7 to 5. When 8 almonds and 8 cashews are removed from the box, the ratio of almonds to cashews becomes 3 to 2. How many cashews were originally in the box?

(A) 40 ✓ (B) 48

(C) 24 (D) 32

1. Write $7k$ almonds and $5k$ cashews, and let x stand for k. After removing 8 of each the ratio is $\dfrac{7x-8}{5x-8} = \dfrac{3}{2}$.
2. Type `y=(7*x-8)/(5*x-8)` and `y=1.5` (zoom out). Tap the intersection: (8, 1.5).
3. Cashews at the start are $5k$: type `5*8` : it returns 40. Check: type `(7*8-8)/(5*8-8)` : it returns 1.5.

Answer: (A)

54. Medium · Desmos

A rectangular banner is 2.4 meters by 1.5 meters. Printing costs \$0.05 per square centimeter. What is the cost, in dollars, to print the banner? (1 meter = 100 centimeters.)

1. The price is per square centimeter, so change the sides to centimeters first: 2.4 m = 240 cm and 1.5 m = 150 cm.
2. Area: type `240*150` : it returns 36,000 square centimeters.
3. Cost: type `36000*0.05` : it returns 1,800 dollars. Slip to avoid: one square meter is $100 \times 100 = 10{,}000$ square centimeters, not 100.

Answer: 1800

55. Medium · Desmos

A train travels at a constant speed of 45 miles per hour. What is the train's speed, in feet per second? (1 mile = 5280 feet)

(A) 3,960

(B) 237,600

(C) 12.5

(D) 66 ✓

1. Feet per hour: type `45*5280` : it returns 237,600.

2. Divide by 3,600 seconds: type `237600/3600` : it returns 66 feet per second.

3. Landmark check: type `88*45/60` : it returns 66, since 60 mph is 88 ft/s.

Answer: (D)

56. Medium · No calculator needed

A rectangular storage box is 0.4 meter long, 0.5 meter wide and 0.3 meter tall. What is the volume of the box, in cubic centimeters? (1 meter = 100 centimeters.)

1. Volume in cubic meters: type `0.4*0.5*0.3` : it returns 0.06.

2. One cubic meter is 100^3 cubic centimeters: type `0.06*100^3` : it returns 60,000 cubic centimeters.

3. Check using centimeters first: type `40*50*30` : it returns 60,000.

Answer: 60000

57. Hard · Desmos

A rectangular banner has width w and length $1.8w$, where $w > 0$. The area of the banner equals the combined area of 4 identical square tiles. The side length of each tile is $k\,w\sqrt{5}$, where k is a positive constant. What is the value of k?

1. The banner's area is $1.8w^2$ and 4 tiles have area $4 \times 5k^2w^2 = 20k^2w^2$. Dividing by w^2, with x standing for k: $20x^2 = 1.8$.

2. Type `y=20*x^2` and `y=1.8` (use the positive crossing). Tap the intersection: (0.3, 1.8).

3. Check: type `4*5*0.3^2` : it returns 1.8, the banner area divided by w^2.

Answer: 0.3